

Stony Brook University
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Fall 2016

ASSIGNMENT 2

Due: Thursday, September 29 in class.

INSTRUCTIONS: Show all your work or points will be deducted.

1. In this problem, we will explore an application of the consumer choice model to show how flexible it is.

Consider the following scenario. You have two midterm exams upcoming, and you have to decide how to allocate your time. After eating, sleeping, exercising and maintaining some level of social life, you have 10 hours each day in which to study for your exams, i.e. 10 hours is your *total amount of study time*. You have figured out that your GPA from your two courses, Intermediate Microeconomic Theory and Biology takes the following form

$$GPA = \frac{4}{7} (2\sqrt{E} + \sqrt{B}),$$

where E is the number of hours you devote to Economics and B are the hours you devote to Biology, which should be regarded as continuous variables. Suppose further that you only care about your GPA.

- (a) Setup the constrained maximization problem and solve for the optimal allocation of study time. You can solve this problem in two ways: You can either setup the Lagrangian¹ or you can use the fact that at an interior optimum, the price ratio should equal the Marginal Rate of Substitution. For the latter method, you need to think about what are the “prices” in this context.
 - (b) If you follow the optimal strategy you found in (a), what will be your GPA?
 - (c) What is the value, measured in GPA units, of increasing your total amount of study time by one more hour? To work this out, you actually need to setup the Lagrangian and solve for λ at the optimum.
 - (d) Now suppose that Intermediate Microeconomic Theory is a very hard class, so for every hour that you study of it, you need to take a one hour break, i.e. if study for E hours Economics, the actual time that you need to spend is $2E$. What is your optimal allocation of study time in this case? What is the value of increasing your total amount of study time?
2. On a given evening, Mr. D.P.² enjoys the consumption of cigars (c) and brandy (b) according to the function.

$$U(c, b) = 20c - c^2 + 18b - 3b^2.$$

- (a) How many cigars and glasses of brandy does he consume during an evening? (Cost is no object to D.P.) *Hint: This is an unconstrained maximization problem.*

¹To be covered in Thursday's Lecture.

²Deep Pockets

- (b) Mr. D.P. recently had a medical check-up in which his doctor advised him that he should limit the sum of glasses of brandy and cigars consumed to 5. How many glasses of brandy and cigars will he consume under these circumstances? *Hint: Notice that D.P. can no longer consume as many glasses of brandy and cigars as in (a).*

3. Mr. Odde Ball enjoys commodities x and y according to the utility function

$$U(x, y) = \sqrt{x^2 + y^2}.$$

- (a) Graph Mr. Ball's indifference curve and its point of tangency with his budget constraint, given by $p_x x + p_y y = I$. What does the graph say about Mr. Ball's behavior? Have you found a true maximum?
- (b) Maximize Mr. Ball's utility if $p_x = \$3$, $p_y = \$4$, and he has \$50 to spend. *Hint: It may be easier here to maximize U^2 rather than U . Why will this not alter your results?*
4. Consider the following utility function (referred as a quasi-linear utility function as it is linear in the second element):

$$u(x, y) = \ln(x) + y,$$

with prices and income given by: $p_x = 1$, $p_y \in \mathbb{R}_+$ and $I \in \mathbb{R}_+$.

- (a) For the special case of $p_y = 2$ and $I = 1$, show analytically and graphically that there is no positive consumption choice $(x^*, y^*) \geq 0$ which satisfies the tangency condition:

$$\frac{u_x(x, y)}{u_y(x, y)} = \frac{p_x}{p_y}.$$

Briefly describe where the failure of the tangency condition stems from and what does it mean. *Hint: Setup the tangency condition and solve it. What do you get? Is it affordable?*

- (b) If we insist that $(x^*, y^*) \geq 0$, then describe, analytically and graphically, the optimal consumption choice for $p_y = 2$, $I = 1$. *Hint: Think about corner solutions.*
- (c) Now suppose that $p_y \in \mathbb{R}_+$ and $I \in \mathbb{R}_+$ are unspecified. Suppose that we *don't insist on positive values* of either x or y . For which values of p_y and I do you get a nonsensical solution in the sense that $y^* < 0$?
5. Mr. A derives utility from martinis (m) in proportion to the number he drinks:

$$U(m) = m.$$

However, Mr. A is very particular about his martinis: He only enjoys them made in the exact proportion of two parts gin (g) to one part vermouth (v). Hence, we can rewrite A's utility function as

$$U(m) = U(g, v) = \min\left(\frac{g}{2}, v\right).$$

- (a) Graph Mr. A's indifference curve in terms of g and v for various levels of utility. Show graphically that regardless of the prices of the two ingredients, Mr. A. will never alter the way he mixes martinis.
- (b) Calculate the demand functions for g and v . Notice that because the problem involves a perfect complements utility function, you cannot solve for utility-maximizing decisions by using calculus.